

- 5 Let k be a constant. The matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are given by

$$\mathbf{A} = \begin{pmatrix} 1 & k & 3 \\ 2 & 1 & 3 \\ 3 & 2 & 5 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & -2 \\ -1 & 3 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{C} = \begin{pmatrix} -2 & -1 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

It is given that $\mathbf{A}$ is singular.

- (a) Show that $\mathbf{CAB} = \begin{pmatrix} 3 & -7 \\ -9 & 3 \end{pmatrix}$. [5]

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- (b) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by $\mathbf{CAB}$. [5]

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- **D** represents an enlargement, centre the origin.
- **E** represents a stretch parallel to the x -axis.
- **F** represents a reflection in the line $y = x$.

[5]

[illegible]